

DEP SPECIFICATION

ROADS, PAVING, SURFACING, CABLE TRENCHES, SLOPE PROTECTION AND FENCING

DEP 34.13.20.31-Gen.

February 2021

ECCN EAR99

DESIGN AND ENGINEERING PRACTICE



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1. INTRODUCTION

1.1 SCOPE

This DEP specifies requirements and gives recommendations for the design and construction of permanent site roads, paved and unpaved areas, civil requirements for cable trenches, erosion protection of slopes, and fencing and gates.

This DEP does not cover construction heavy-haul roads or mining roads.

The design of contaminated and non-contaminated drainage systems is not covered by this DEP. The design of drainage systems is described in DEP 34.14.20.31-Gen.

The site preparation work, including preparation of sub-grade and installation of subbase, are not within the scope of this DEP and instead are covered in DEP 34.11.00.11-Gen.

Site specific layout strategy for the road layout requirements are covered in DEP 80.00.10.11-Gen..

For additional electrical requirements for cable trenches, refer to DEP 33.64.10.33-Gen. and either DEP 33.64.10.10-Gen. (outside North America) or DEP 33.64.20.10-Gen. (within North America), whichever is applicable.

This DEP is a revision of the DEP of the same title and number dated February 2017; see (1.5) regarding the changes.

1.2 DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS

Unless otherwise authorised by Shell GSI, the distribution of this DEP is confined to Shell companies and, where necessary, to Contractors and Manufacturers/Suppliers nominated by them. Any authorised access to DEPs does not for that reason constitute an authorisation to any documents, data or information to which the DEPs may refer.

This DEP is intended primarily for use in facilities related to oil refineries, chemical plants, gas plants, or any such facility related to manufacturing activities. Where applicable, it may be used in exploration, production and distribution.

When DEPs are applied, a Management of Change (MOC) process shall be implemented; this is of particular importance when existing facilities are to be modified.

If national and/or local regulations exist in which some of the requirements could be more stringent than in this DEP, the Contractor shall determine by careful scrutiny which of the requirements are the more stringent and which combination of requirements will be acceptable with regards to the safety, environmental, economic and legal aspects. In all cases, the Contractor shall inform the Principal of any deviation from the requirements of this DEP which is considered to be necessary in order to comply with national and/or local regulations. The Principal may then negotiate with the Authorities concerned, the objective being to obtain agreement to follow this DEP as closely as possible.

1.3 DEFINITIONS

1.3.1 General definitions

The **Contractor** is the party that carries out all or part of the design, engineering, procurement, construction, commissioning or management of a project or operation of a facility. The Principal may undertake all or part of the duties of the Contractor.

The **Manufacturer/Supplier** is the party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.

The **Principal** is the party that initiates the project and ultimately pays for it. The Principal may also include an agent or consultant authorised to act for, and on behalf of, the Principal.

The word **shall** indicates a requirement.

The word **should** indicates a recommendation.

The word **may** indicates a permitted option.

1.3.2 Specific definitions

The following definitions are specific to this DEP.

Term	Definition
Base	The layer or layers of specified or selected material of designed thickness placed on a subbase or subgrade to support surface.
Battery limit	The operational boundary of individual processing or utility units.
California Bearing Ratio	The California Bearing Ratio test is a penetration test used to evaluate the subgrade strength of roads and pavements. Results of test are used to determine thickness of flexible pavement and its component layers.
Channel	Open flow drainage component that is typically trapezoidal in cross section and are without continuous covers (either solid or grated). The interior finish (surface exposed to the flow contents) of a channel depends on the contents, the conditions of flow and location. The word channel is synonymous with ditch or swale.
Construction Joint	Construction joints are stopping places in the process of construction.
Construction Roads	Roads that provide greater than typical load capacity designed for heavy haul truck traffic that may require a greater than typical radius and lane width. Certain construction roads transition from heavy haul vehicles transporting equipment and materials for construction to operations and maintenance.
Contraction Joints	Contraction joints, also known as control joints, are placed to control random cracking, and required depending upon slab thickness and spacing of construction joints.
Expansion Joints	Expansion joints, also known as isolation joints, are used to relieve flexural stresses due to vertical movement of slab-on-grade applications that adjoin fixed foundation elements such as columns, building or machinery foundations. This allows for thermal expansion and contraction without inducing stress into the system.
Flexible Paving	Flexible paving is composed of asphaltic concrete.
Impermeable	The design materials need to provide adequate barrier for incidental exposure to a specific concentration of the leaked material for a specified duration per local code requirements.
Main Roads	Roads that access off-plot areas, on-plot areas, and the primary gate entrances are typically considered main roads. Main roads provide access to product shipping points and receiving points, and provide space for construction and maintenance vehicles. Main roads are typically used by large trucks and cranes required for construction and maintenance. Certain main roads are expected to manage large loads over the life of the facility and might require a greater than typical radius and lane width.

Term	Definition
Maintenance Roads	Maintenance roads provide access to product shipping/receiving points and provide space for maintenance vehicles. Maintenance roads are typically used by large trucks, cranes, and vehicles required for maintenance and turnarounds, and provide access to equipment within on-plot areas.
Off-plot areas	An area designated for all facilities not in the on-plot area which can include administration buildings, workshops, laundries, warehouses, materials yards, storage tank compounds, pump stations, flares, fire-fighting station, training grounds, cooling water intake station, cooling towers, cooling water settling ponds, cooling water discharge channels, jetties, harbour, associated pipe tracks and roads.
Off-site	Areas associated with the site, but located at a distance, e.g., water wells and well pads.
On-plot area	An area designated for utility and processing units, including associated control rooms, electrical sub-stations, analyser houses, stacks, associated pipe tracks and plant roads.
Patrol Roads	Patrol type vehicles weight about 5000 kg (11 kips) and can handle steeper inclines and are at times 4-wheel drive rendering the need for the roadway specific requirements void. The need for Patrol roads is typically specified in the Site Security Assessment.
Paving	Cementitious or asphaltic concrete that distributes the load to base and subbase. Synonymous to paved surface.
Plant road	A road within in an on-plot area. Plant roads provide access to equipment within on-plot areas by maintenance vehicles and personnel vehicles.
Rigid Paving	Rigid paving is composed of Portland cement concrete which has low flexibility and distributes wheel loading over a wide area of subgrade.
Road width	The distance between the outside edges of the road pavement (i.e. vehicular traffic lanes). This does not include the width of shoulders.
Rough grade	the vertical elevation location of the soil ground surface that approximately conforms to the approved finished soil grade elevation plan.
Site	An area comprising the total of all on-plot and off-plot areas, which are located as one operational entity.
Subbase	A compacted layer of graded fill provided over prepared subgrade soils to improve drainage, to resist frost heave, to provide structural support by distributing loads, and to prevent pumping of finer grained soils into base material at paving joints and edges.
Subgrade	Foundation soil placed and compacted during initial grading work. It consists of natural soils found on site, unless soil strength is inadequate and imported soils are required.
Surface	One or more layers of a pavement structure designed to accommodate traffic load, the top layer of which resists skidding, traffic abrasion and the disintegrated effects of climate and environment.

1.3.3 Abbreviations

The following abbreviations are specific to this DEP.

Term	Definition
ACI	American Concrete Institute
CBR	California Bearing Ratio

1.4 CROSS-REFERENCES

Where cross references to other parts of this DEP are made, the referenced section or clause number is shown in brackets (). Other documents referenced in this DEP are listed in (7).

1.5 SUMMARY OF MAIN CHANGES

This DEP is a revision of the DEP of the same number dated February 2017. The changes followed a requirements quality review and were not a full technical update. The following are the main, non-editorial changes.

Section/Clause	Change
2.2	Added requirements to cover access to public roads and shipping/receiving points.
2.3	Added testing options to determine subgrade and subbase strengths
2.4.4	Replaced the former Table 2 with a reference to AASHTO GDHS
4	Updated cable trench requirements to align with electrical DEPs

1.6 COMMENTS ON THIS DEP

Comments on this DEP may be submitted to the Administrator using one of the following options:

Shell DEPs Online (Users with access to Shell DEPs Online)	Enter the Shell DEPs Online system at https://www.shelldeps.com Select a DEP and then go to the details screen for that DEP. Click on the "Give feedback" link, fill in the online form and submit.
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DEP Standard Form (other users)	Use DEP Standard Form 00.00.05.80-Gen. to record feedback and email the form to the Administrator at standards@shell.com .

Feedback that has been registered in the DEP Feedback System by using one of the above options will be reviewed by the DEP Custodian for potential improvements to the DEP.

1.7 DUAL UNITS

This DEP contains both the International System (SI) units, as well as the corresponding US Customary (USC) units, which are given following the SI units in brackets. When agreed by the Principal, the indicated USC values or units may be used.

1.8 NON NORMATIVE TEXT (COMMENTARY)

Text shown in italic style in this DEP indicates text that is non-normative and is provided as explanation or background information only.

Non-normative text is normally indented slightly to the right of the relevant DEP clause.

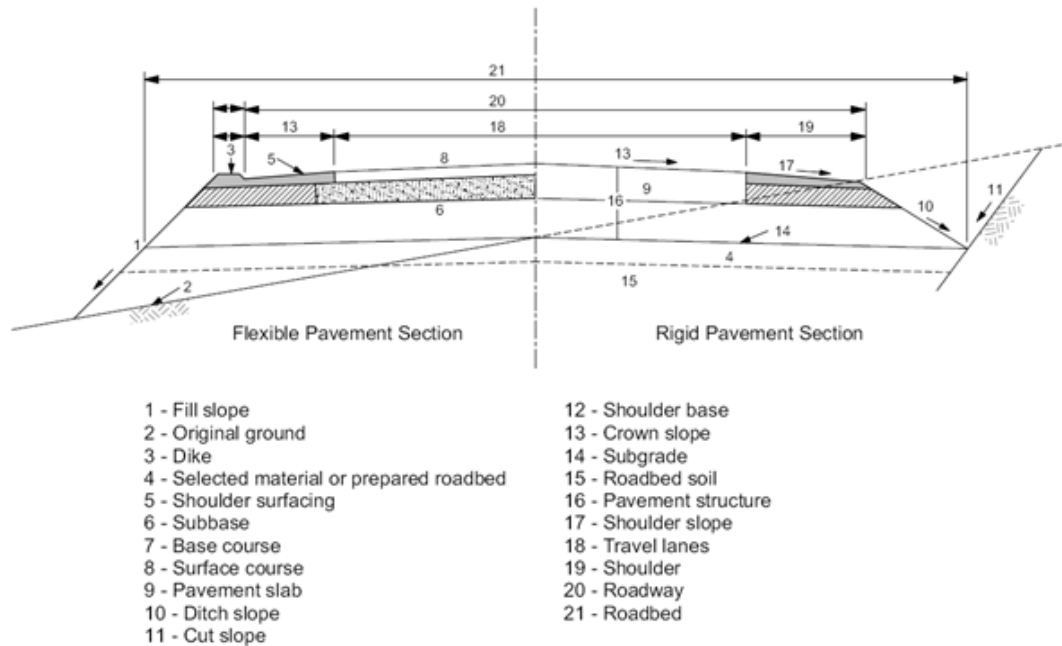
2. ROADS

2.1 CLASSIFICATION

2.1.1 General

1. Refer to Figure 1 for the illustration of finished surface nomenclature.

Figure 1 Schematic drawing showing example of cross-sections of roads



2. Roads shall be classified based on the function (i.e. location of the road or pavement, and intended use), load duty and surface type.

Classification definitions are given in (2.1.2) through (2.1.4).

Rationale: The classification determines the required width, turning curves, horizontal and vertical gradients, minimum number of lanes, shoulder requirements and design speeds.

2.1.2 Functional classification

1. The functional classification determines the required width, turning curves, horizontal and vertical gradients, minimum number of lanes, shoulder requirements, and design speeds. The following functional classifications are adopted based on location of the road or pavement and intended use:

- o Main roads
- o Off plot roads
- o Plant roads
- o Maintenance roads
- o Patrol roads
- o Construction roads

2.1.3 Load duty classification

1. The following load duty classifications shall be adopted:

- a. Light;
- b. Heavy.

Rationale: The load duty classification determines the number and location of wheels and load distribution, maximum allowable settlement, maximum crack width, design life, performance criteria and traffic load combinations including number of axle movements. Study the specific traffic requirements of all area roads, and specify the road requirements accordingly.

2. The following roads shall be classified as heavy duty:

- a. main roads on site, roads leading to the main buildings, main process units, utility areas, loading facilities, distribution areas and warehouse areas;
- b. main access roads to the site whose construction and maintenance are not the responsibility of government authorities.

Rationale: Heavy duty roads provide access to product shipping/receiving points, so they require space for maintenance vehicles to pass. Heavy duty roads are typically used by large trucks and cranes, therefore qualifying as heavy duty. All other roads are classified as light duty.

2.1.4 Surface type classification

1. The following surface type classifications may be adopted:

- a. flexible pavement;
- b. rigid pavement;
- c. concrete block pavement;
- d. gravelled;
- e. unpaved.

2.2 ROAD SYSTEM DESIGN

1. All components and elements of a road shall be designed and constructed to be fit for purpose during the lifetime of the site (2.5.1).
2. Roads shall provide stable and durable access for each of the following:
 - a. to all locations of the site which require vehicle transport means;
This includes cars, trucks, cranes and other mobile equipment such as snow ploughs, fire engines or road sweepers that are required during the lifetime of the plant.
 - b. for heavy vehicles transporting equipment and materials during construction, maintenance or turnarounds.
3. Access to public street and highway systems shall be included in the design of a new site, or construction of a project at an existing site.
4. Depending on which public entity has jurisdiction at the access point, local, state, or federal regulations shall be consulted for design of alignment, acceleration and deceleration velocities, signage, and roadway markings.
5. Shipping and receiving points of products, supplies or feed stocks by truck shall be included in the roadway system design.

2.3 SUBBASE AND SUBGRADE

1. The classification of subgrade soil shall be determined for the road pavement design.
2. The geotechnical contractor shall perform the geotechnical soil investigation and field tests required to obtain: soil physical properties (e.g. soil classification, density, water content), stiffness and/or strength (e.g. resilient modulus (M_R), modulus of subgrade reaction (k), California Bearing Ratio (CBR), and Resistance Value (R)), thermo-hydraulic properties (e.g. drainage coefficients, permeability, coefficient of thermal expansion).
3. The geotechnical contractor shall provide local recommendations for the stabilization of in situ soils, replacement of in situ soils with suitable borrow materials, economical specifications for subgrade, subbase, and locally available roads construction materials.
4. The following tests may be used to determine subgrade and subbase strengths:
 - a. Plate bearing test per ASTM D1195 – used for subgrade evaluations, strength measurements for subbase, or finished paving;
 - b. California Bearing Ratio (CBR) test method per ASTM D1883 – used to evaluate the potential strength of subgrade, subbase, and base course materials, including recycled materials for use in design of road paving. The CBR value obtained in this test forms is an integral part of several flexible pavement design methods;
The CBR value of the sub-grade is recommended to be equal to or greater than 10, to permit typical economic designs of paved surfaces that function to support maintenance or operational type traffic loads in roads.
 - c. Resistance value method per ASTM D2844 – used to measure the strength of subgrade, subbase, and base course materials for use in road paving. The R-value is used as criteria for acceptance of aggregate for base course and bituminous courses.
5. The design shall include measures to prevent intermixing of subgrade material into the subbase during construction and during the design life of the pavement.

2.4 GEOMETRICAL DESIGN

2.4.1 Lateral gradient and cross fall

1. All roads shall have a cross fall (i.e., cross slope) from the centre or crown of the road.
 - a. If the highest road elevation is at the centre or crown of the road, then a ditch should be located on both sides of the road.
Occasionally a cross fall can be continuous across the road. This could be a cost effective solution, draining into one ditch.
2. The cross fall shall have a gradient of 1 in 50 to ensure effective run-off of rainwater.
3. The paved areas adjacent to the roads shall have a smooth transition to the edge of the roads.
4. The road maximum gradient (slope) in the longitudinal direction shall be 1 in 20 for heavy-duty roads and 1 in 10 for light-duty roads.

2.4.2 Cross section

1. All roads, except maintenance roads and patrol roads, shall be of the two-lane type.
2. Road widths shall be based on service requirements during the life of the plant, including construction, maintenance, operation and turnaround/overhaul.
3. Road widths shall comply with the minimum requirements of Table 1.

Table 1 Minimum road width requirements

Road Type	Minimum Road Width (m)	Minimum Road Width (ft)	Notes
Main road	8	26	shoulder
Offplot road	6	20	shoulder
Plant road	4	13	shoulder
Patrol road	3	10	no shoulder
Maintenance road	to suit	to suit	no shoulder
Construction road	to suit	to suit	no shoulder

4. The section of road at fire hydrant locations shall have one of the following:
 - a. a minimum width of 7 m (23 ft) for a length of 15 m (50 ft) to allow fire vehicles to be parked clear of traffic;
 - b. turnouts for roadway sections that do not meet these minimum dimensions.
5. Length of turnouts shall be a minimum width of 7 m (23 ft) and a minimum length of 15 m (50 ft) to allow fire vehicles to be parked clear of main traffic lanes.
6. Shoulders shall be capable of supporting, without damage, slow-moving and parked vehicles for which the road is designed.
7. Sufficient shoulder width shall be provided to accommodate cable trenches.
8. The road (i.e., subbase and pavement) shall be adequately supported horizontally either by a sufficiently paved shoulder or by a vertical retaining element, e.g., concrete kerb.
9. The boundary between the road and road shoulder shall be clearly marked with lines or a marker or a kerb.
10. Openings in lines or kerbs shall ensure the drainage of the road surface.
11. Road shoulders on either side shall have a minimum width of 2 m (6.5 ft) for heavy-duty loads and 1.5 m (5 ft) for light-duty roads.

2.4.3 Horizontal and vertical curves

1. Roads shall have a vertical curve length of 30 m (98.5 ft) when the difference in slope of the road gradients is 2% or greater.

Rationale: The center line of a road consists of a series of straight lines. They are interconnected by curves that are used to change alignment, direction, or slope of the road. The curves that change the alignment or direction are known as horizontal curves, and those that change the slope are vertical curves.
2. Design speeds shall be based on the function and location of the road.

2.4.4 Road crossings

1. Road intersections shall cross at an angle of 90 degrees.
2. At road crossings, the minimum turning radius, also known as radius of curvature, shall be based on turning requirements of vehicles that are expected to use the road regularly.

2.5 ROAD PAVEMENT DESIGN REQUIREMENTS**2.5.1 Design lifetime**

1. Pavement for roads, including turn out and parking areas shall be designed for the lifetimes, also known as design period, in Table 2.

Table 2 Design lifetimes

Design	Lifetime
Asphalt	20 years
Concrete	30 years

2.5.2 Performance criteria

1. The Present Serviceability Index (PSI) or terminal serviceability index (p_t) for heavy duty roads and for light duty roads shall be 2.5 as defined in the AASHTO GDPS-4 publication.

Rationale: Selection of the lowest allowable present serviceability index or terminal serviceability index is based on the lowest index that will be tolerated before rehabilitation, resurfacing, or reconstruction becomes necessary.

2. When designing road pavement, the parameters in Table 3 and Table 4 shall be used.

Table 3 Performance criteria (Serviceability)

Standard deviation for traffic for flexible pavements	$S_o = 0.45$
Standard deviation for traffic for rigid pavements	$S_o = 0.35$
Initial serviceability index for flexible pavements	$p_o = 4.2$
Initial serviceability index for rigid pavements	$p_o = 4.5$

NOTE: Refer to AASHTO GDPS Part I Chapter 4.

Table 4 Performance criteria (Reliability)

Classification	Reliability (R)	Standard Normal Deviate (Z_R)
heavy duty roads	75 %	-0.674
light duty roads	50 %	0.000
Patrol	50 %	0.000
maintenance roads	N/A	N/A

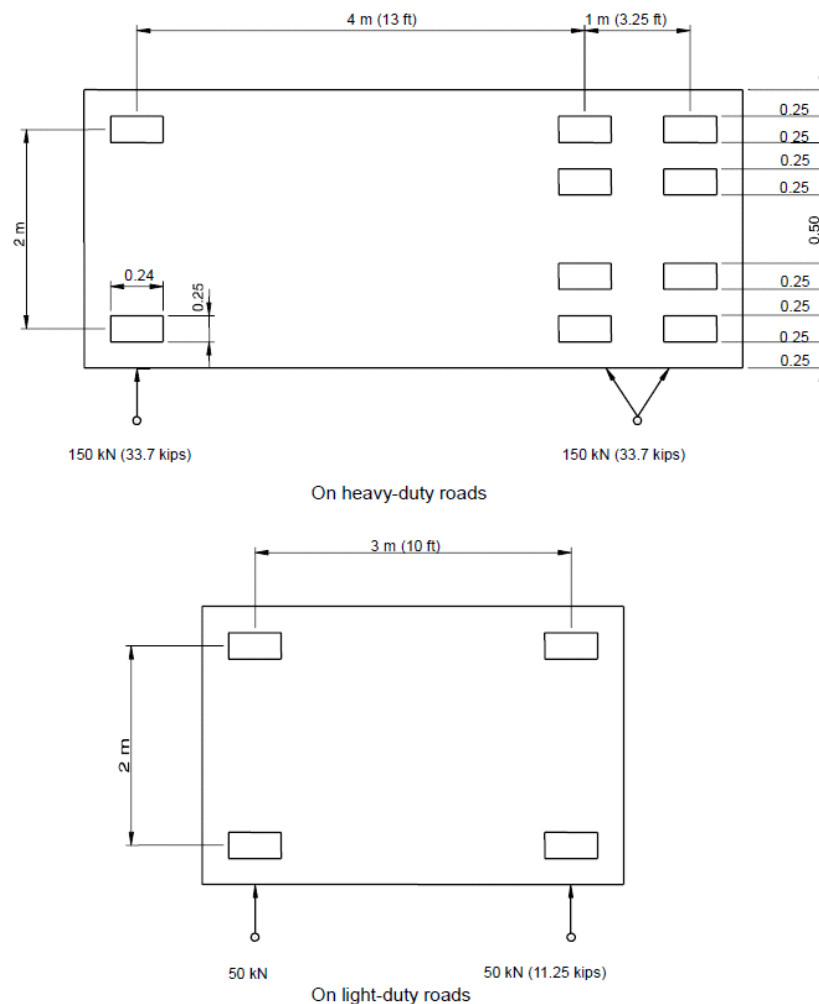
NOTE: Refer to AASHTO GDPS Part I Chapter 4.

2.5.3 Design Requirements, Standards and Manuals

Information required for the design of road pavement includes vehicles types, average daily volume of traffic anticipated during the design life, wheel loads, including the maximum single-axle and tandem-axle loading for trucks, forklifts, mobile work platforms, and cranes. The axle loads and arrangement of vehicle dimensions for heavy- and light-duty roads for a project that is not specified to meet a specific AASHTO or other local country loading requirement are defined in Figure 2.

1. Road pavement design and construction shall be in accordance with one of the following:
 - a. AASHTO Guide for Design of Pavement Structures (AASHTO GDPS);
 - b. Design Manual for Roads and Bridges of the Highway Agency (UK);
 - c. local country highway department specifications requirements.

Figure 2 Schematic drawing showing axle loads and vehicle dimensions



2. The road pavement design and construction shall make use of the specified low speed of the traffic and the impact of the static loading of the trucks and other equipment.
3. For a project that is not specified to meet a specific AASHTO or other local country loading, traffic loading shall be specified based on the recommended number of axle movements as in Table 5.

Table 5 Traffic Loading

Road	Loading
heavy-duty roads	1.5 million
light-duty roads	1.0 million

2.5.4 Exceptional transport

1. The most severe conditions, as specified, shall be adopted as the extreme loading conditions in the design of roads (including culverts and bridges).
2. At least one road leading to the main process or distribution areas shall be designated as a heavy duty road.

2.5.5 Patrol roads

1. Patrol roads shall be designed to accommodate vehicles with a maximum weight of 5,000 kg (11,000 lb) under all relevant climatic conditions.
2. The maximum gradient of patrol roads shall not exceed 1 in 8.5.
3. Patrol roads shall be paved with a flexible type paving.

2.5.6 Maintenance roads

1. Maintenance roads shall facilitate access to areas within the site boundary for 4-wheel drive type vehicles under normal weather conditions.
2. Vegetation, peat and soft clays shall be removed.
3. The maximum gradient of maintenance roads shall not exceed 1 in 5.

Surfacing of maintenance roads is not required.

2.6 DRAINAGE**2.6.1 Surface drainage**

Roads drainage systems consist of collection and removal of surface water runoff from roads and shoulders. Culverts, ditches, swales, kerbs, and gutters are parts of the roadway drainage system.

Refer to (2.4.1) for lateral gradient and cross fall (cross slopes) requirements.

1. Refer to DEP 34.14.20.31-Gen. for the requirements for surface drainage.

The surface paving forms an integral part of the drainage system

2.6.2 Control ground water

1. The ground under the road system shall be drained.

Subgrade sloped to ditches is normally used to provide subsurface ground water drainage. If ditches are not appropriate, subdrains can be used.

Rationale: To prevent frost issues, and to maintain a layer of stable material to distribute pavement pressure loads.

2. The distance between the underside of the paving and the normal high ground water level shall be not less than the distances in Table 6.

Table 6 Distance from high ground water level to bottom of paving

Road Type	Distance to water (m)	Distance to water (ft)
Main road	0.75	2.5 ft
Off plot road	0.50	1.67 ft
Plant road	0.50	1.67 ft
Patrol road	0.30	1 ft
Maintenance road	not applicable	not applicable

3. If the subgrade comprises of rock, the normal high ground water level shall be no higher than the subgrade level.
4. A permanent sub-surface drainage system shall be designed to ensure control of the ground water level by using the minimum distances specified in Table 6.
5. A permanent sub-surface drainage system shall include the drainage of the subsoil under and adjacent to the road.

Rationale: The drainage under the roadway is typically away from the shoulder of the road. In some cases, a channel is provided so that water is managed adjacent to the shoulder.

6. The influence of unlined drainage ditches, lined drain ditches and channels with weep holes on the ground water table shall be included in the design of roadside drainage systems.
7. If the drainage system is lined, the ground water levels beneath the road shall be controlled by an independent system.
A lined drainage system can include a buried drainage blanket and central drain.
8. The drainage system design shall prevent the migration of fine soils into the drainage layers or surface ditches.

2.7 BRIDGES, CULVERTS AND PIPE CROSSINGS

In this DEP, design considerations for bridges, culverts, pipe crossings and underground facilities underneath roads are limited to those that interface with road design.

1. Bridges, culverts and other underground facilities shall be designed for the dead loads (i.e. earth loads based on unit weight of soil times depth of cover), live loads, such as maximum expected loading condition, as indicated in (2.5.3) and (2.5.4).
Culverts are used to channel water under roadways and driveways or to remove drainage from ditches.
2. If the construction of roads interferes with the existing surface drainage system, the road design shall ensure the existing drainage system is unaffected.
Measures taken to achieve can include the installation of culverts and ditches and the re-routing of water courses
3. In order to avoid damage to underground facilities underneath a road, the minimum soil cover depth (2.7.4) shall be based on rough grade elevation of soil, if underground facilities will be subject to heavy construction loads before road paving is installed.
4. The minimum soil cover depth (i.e., the vertical distance between the rough grade elevation of soil and the top of underground facilities) may vary, and shall be defined per local soil characteristics and geotechnical engineering investigation report recommendations.

5. The length of culvert or bridge underneath a road shall not be less than the road width including shoulder steps where provided, side slopes, and a safety margin of 1.5 m (5 ft).
6. The concrete culvert or bridge underneath a road shall have a minimum cross section width of 460 mm (18 in), and a minimum depth of 460 mm (18 in).
7. For open concrete culverts, the minimum clearance above the top of the piping to the underside of the roof in culverts shall be 500 mm (20 in) after accounting for settlement.
8. Piping under the bridge shall be accessible for maintenance and inspection.

2.8 BARRIERS

2.8.1 Crash barriers

1. Crash barriers shall be applied where specified to protect personnel, equipment and piping.
2. Crash barriers shall be demonstrated to be designed adequately to protect equipment and personnel in potential collisions.
3. The system and material selection for the crash barriers shall be designed to absorb impact energy by deformation without collapse.

2.8.2 Road barriers

1. Road barriers at site entry points shall be of the vertical rotating type (drop arm barrier), either with counterweight or with integrated gas compression.

Rationale: Road barriers form a hard barrier to vehicle entry and are typically used in conjunction with barrier arms.

2. The material of the barrier arms shall allow for deformation upon impact or be provided with breakaway pins.

Rationale: Barrier arms assets have a history of frequent failures due to inadvertent vehicle contact. This requirement allows for ease in repair.

3. Barrier arms, road barrier and their control systems shall permit access by emergency teams.
4. Barrier arms shall be fully painted in red and white reflecting bands.

Rationale: For ease of identification for vehicles.

5. The plant road and the barrier shall be lit by lampposts.
 - a. Refer to Appendix C of DEP 33.64.10.10-Gen. for the required illumination levels.

2.9 TRANSITION SLABS

1. A reinforced concrete transition slabs should be used in locations where differential settlement between paving and foundations is expected.

This can be the case at pipe culverts or piled objects such as foundations and liquid-tight process plant paving.

2.10 MATERIALS

1. The quality of the construction works, materials, finishes of structures, external works and testing procedures shall match international quality standards listed in (7).
2. Before surfacing, acceptance of the sub-grade and subbase shall be obtained through compliance testing.
3. The results of compliance testing shall meet the requirements of the acceptance criteria specified in (2.5.2).

4. Dust control measures during construction of plant roads, patrol and maintenance roads shall be implemented.

Rationale: Dust control measures might include the use of co-polymer emulsion, calcium chloride or lime/cement stabilization.

3. SURFACED PLANT AREAS

3.1 GENERAL

1. The selection of pavement type shall be based on the expected loads and the local conditions, including subsoil.

3.2 SURFACE SYSTEM DESIGN

1. Surfaced plant areas paving selection shall include the following functional requirements:
 - a. allow access of personnel and equipment for construction, operation and maintenance;
 - b. a foundation base for light equipment, e.g. sheds, racks, ladders, stairs, miscellaneous pipe supports, and platforms;
 - c. protection of the soil and the ground water from contamination resulting from a spillage of chemicals or hydrocarbons;
 - d. prevention of erosion of the soil specified in (5);
 - e. routing of spillages to sumps and drains as specified in (3.5).

3.3 SUBBASE AND SUBGRADE

1. Refer to (2.3) for sub-grade and subbase requirements.

A modulus of sub-grade reaction is used to represent all soil supporting the concrete paving slabs. The pavement can rest directly on the sub-grade prepared during the site preparation. A subbase to support the paving might be required depending on the suitability of the sub-grade per geotechnical recommendations.

3.4 PAVEMENT DESIGN REQUIREMENTS

3.4.1 Design lifetime

1. Refer to (2.5.1) for requirements related to design lifetime.

3.4.2 Design standards and manuals

For paving accessible to traffic, the design requirements, standards and manuals are covered in (2.5.2) and (2.5.3).

1. For paving not accessible for traffic, minimum loads of 10 kN (2250 lbf) concentrated load and 5 kN/m² (0.725 lbf/in²) uniform distributed live loads shall be used.

3.4.3 Joints in concrete pavement

1. Reinforced concrete paved areas shall be divided into slabs separated by flexible expansion joints, contraction joints, or construction joints, to allow for lateral movements in accordance with Joint Design for Concrete Highway and Street Pavements Issued by Portland Cement Association (PCA), or UFC 3-320-06a Concrete Floor Slabs on Grade Subject to Heavy Loads, or equivalent local requirements:
 - a. Refer to ACI 325.9R Guide for Construction of Concrete Pavements for construction requirements.
2. All joints in accidentally contaminated and continuously contaminated areas as per DEP 34.14.20.31-Gen. shall be designed to meet local regulatory requirements.

3. Concrete slabs shall have expansion joints with joint material where differential movement is predicted between the pavement and other adjacent vertical surfaces (e.g., walls, columns, catch basins, buildings, manholes and equipment foundations).
4. The locations of lateral movement joints shall be selected together with the locations of slab reinforcements to prevent uncontrolled cracking as a result of shrinkage, thermal variations, thickness or horizontal dimension variations, natural changes in moisture content and/or other conditions.

Rationale: This is to prevent uncontrolled cracking as a result of shrinkage, thermal variations, thickness or horizontal dimension variations, natural changes in moisture content and other conditions.

5. All concrete pavements for heavy-duty service shall be constructed using a thickened slab.

3.4.4 Concrete kerbs

1. Concrete kerbs shall be provided to accomplish all of the following:
 - a. to separate vehicular traffic from pedestrian areas;
 - b. to channel surface water runoff;
 - c. to confine hydrocarbon or chemical spills and with impermeability criteria specified in (3.4.2).

3.5 DRAINAGE

1. Refer to DEP 34.14.20.31-Gen. for the drainage system requirements.

The pavement forms an integral part of the drainage system.

3.6 CONCRETE BLOCK PAVEMENT

1. Concrete block pavement shall be installed in stretcher bond or herringbone pattern.
2. Concrete blocks shall be manufactured according to EN 1338 or according to equivalent standard.
3. Design of the concrete block pavement shall be made in accordance with the reference 'Port and Industrial Pavement Design with Concrete Pavers'.
4. If there are level differences between adjacent surfaces that have block paved surface, then the transition that has linear level differences shall relate to a maximum slope of 1:20 and in the transverse direction with a maximum slope of 1:40.

3.7 UNPAVED AREAS

1. Surfacing of unpaved areas shall prevent soil erosion by wind and water.
2. Surfacing of unpaved areas shall be able to support pedestrians and light vehicles, including 4-wheel drive vehicles (2,500 kg to 3,000 kg (5,500 lb to 6,600 lb) maximum).
3. Surfacing of unpaved areas shall minimize the growth of vegetation.
4. Unpaved areas should be classified into two categories as follows:
 - a. low fire hazard, which include the following areas
 - i. neutral zone between fences;
 - ii. unused off-plot areas;
 - iii. off-plot pipe racks and pipe tracks containing continuously welded piping with no flanged connections.

- b. fire hazard, which include the following areas:
 - i. flare and open fire areas;
 - ii. off-plot pipe racks and pipe tracks which contain flanged connections, valves or sampling points, and include manifolds;
 - iii. areas around processing units.
- 5. Pipe tracks shall have a minimum 75 mm (3 in) thick gravel cover.
- 6. In all on-plot areas, gravel, or other inert material shall be in a layer thickness of at least 50 mm (2 in).
- 7. Off-plot areas shall be covered with grass.

3.8 MATERIALS

- 1. The design and construction of concrete, paving shall be in accordance with DEP 34.19.20.31-Gen.

Where gravel is used in paving, crushed, sound rock and coarse gravel are generally acceptable.
- 2. Surface materials shall be able to withstand local environmental conditions.
- 3. Surface materials shall be inert or self-extinguishing in the events of fire.

4. CABLE TRENCHES

This section is limited to general civil engineering minimum requirements only.

- 1. Cables laid in cable trenches shall be at least 600 mm (24 in) below ground level.

The depth of the trench can be increased in order to accommodate up to 3 layers of cables.
- 2. Except for short lengths near the termination points of the cables in the plant area, underground ducts shall not be used.

Rationale: Since relatively long ducts will affect the cable current rating unfavourably.
- 3. Cable trenches shall be filled with graded, non-angular, well-draining, compacted sand with the following properties:
 - a. a maximum size of 2 mm particle passing a number 8 sieve (BS EN) or number 10 sieve (ASTM);
 - b. free of sharp particles.
- 4. At locations where sub-zero temperatures are expected the effects of frost shall be included in the design depth of backfill materials for cable trenches.
- 5. The location of cable trenches shall be indicated on the surface in compliance with site standards.
- 6. All trenches shall be covered with removable concrete panels that meet the same standard as the surrounding paving and that can be removed with available site equipment.
- 7. Trenches for electrical cables shall be covered by red concrete paving.
- 8. Trenches for instrumentation cables shall be covered by green concrete paving.
- 9. In order to facilitate cable laying before the concrete paving is finished, the expansion seams in concrete floors may be located on either side of the cable trenches.
- 10. If permanently covered cable trenches are used, sand filled dummy trenches, 0.6 m to 1 m (2 ft to 3 ft) wide, and covered by concrete paving, shall be provided for future cabling.

11. The Contractor shall develop drawings for cable trenches in concrete paving.
 - a. Refer to Standard Drawings S 19.001 as a guide to develop project specific standard drawings for cable trenches in concrete paving.
 - b. Refer to Standard Drawings S 19.002 as a guide to develop project specific standard drawings for cable trenches in unpaved, brick-paved or tiled areas and crossing roads.
12. At the end of concrete cable trenches, short ducts or pipes shall slope down into the surrounding soil, to prevent cable damage at the edges owing the settling of the soil.
13. Where cable trenches cross roads, an additional 20% of the total number of ducts shall be provided at the normal cable laying depth to accommodate future cables.
If 1 to 5 ducts are present, then one duct is added. If 6 to 10 ducts are present, then two ducts are added. The size of the trench generally correlates with the number of ducts present.
14. For locations where cable trenches intersect a bridge for the upper trench, comprising a concrete bottom slab supported on the walls of the lower trench, shall be constructed.
Rationale: To facilitate excavation of the lower cable trench.
15. Cables shall have a minimum distance of 0.6 m (2 ft) to any buried piping or pipelines.
Rationale: To prevent mechanical or temperature damage to cables.
16. Cables shall cross underneath buried piping or pipelines except where the depth measured at the top of the piping or pipeline is more than 1 m (3 ft).
Rationale: In order to prevent mechanical or temperature damage to cables.
17. To allow efficient design and installation of underground utilities at the beginning of the work, and underground elevation protocol that specifies depth and installation sequence of underground utilities, including cables, shall be jointly reviewed by the Civil and Electrical Contractor engineers.

5. EROSION PROTECTION OF SLOPES INCLUDING, EMBANKMENTS, DITCHES AND OPEN DRAINS

5.1 GENERAL

The scope of this section is limited to erosion protection requirements.

The stability, settlement behaviour and construction of slopes, bund walls and embankments are covered in DEP 34.11.00.11-Gen. and DEP 34.11.00.12-Gen.

For ditches and open drains, only the material, design, and construction aspects of the protective surface layer are considered.

Design and capacity requirements for drainage systems are covered in DEP 34.14.20.31-Gen.

1. A filter fabric shall be applied to prevent erosion e.g., through joints in brickwork and slabs, cracks in the protection layer.

5.2 DESIGN

5.2.1 General

1. Slopes for bund walls, cuts, embankments, open drains, channels and ditches shall be protected against erosion and subsequent damage, and against failure as a consequence of settlement, wind, water, flow and spillages for the life of the facility.

Rationale: The purpose of erosion protection is to prevent the process of weathering and transport of solids (sediment, soil, rock and other particles).

2. Selection of slope protection shall include activities that can occur in the close vicinity of the slope and could have a detrimental effect, e.g., temporary loading from vehicles, storage of materials, or vibration.
3. The design of the slope protection shall include long term settlement, erosion or other behavioural processes.

Rationale: To ensure that the slope protection system remains stable during its design life.

4. Where impermeable erosion protection systems are selected to prevent contamination of ground and groundwater, the design of the erosion protection system shall not include drainage points or weep holes.
 - a. The design shall include the highest possible ground water level in combination with the minimum water level in the drain to design against slope failures due to build-up of pore water pressures.
5. For erosion protected areas that can accidentally be contaminated as per DEP 34.14.20.31-Gen. the cover layer of open drains and ditches shall protect the subsoil against pollution.

5.2.2 On-plot areas

5.2.2.1 Inside battery limits

1. Inside battery limits, all slopes, bund walls, embankments, open drains, and ditches shall be lined with concrete.

Rationale: To ensure that these systems remain stable during the design life.

2. Refer to DEP 34.19.20.31-Gen. for the requirements for the concrete.
3. The minimum layer thickness of the lining concrete shall be 100 mm (4 in).

Rationale: A reinforcement mesh of 7 mm (1/4 in) steel wire spaced at a pitch of 200 mm x 200 mm (8 in x 8 in) is the typical used with this thickness of concrete.

4. A reinforcement mesh of 7 mm (1/4 in) steel wire spaced at a pitch of 200 mm x 200 mm (8 in x 8 in) shall be incorporated into the design of the concrete lining.

Rationale: The reinforcement provides additional stability.

5. Concrete slabs shall have a maximum size of 8 m x 8 m (26 ft x 26 ft).

Rationale: To control cracking due to thermal effects like curling and drag. The slabs are typically divided and separated by flexible expansion or construction joints to allow for lateral movement.

6. Slopes higher than 1 m (3.25 ft) shall be provided with drainage points or weep holes to eliminate the build up of pore water pressures.
 - a. Wash out of soil through drainage points or weep holes shall be prevented.
7. Erosion of soil through joints between panels shall be prevented, e.g., by the application of strips of filter material.
8. Long-term behaviour, e.g., danger of gradual clogging, shall be factored into the design.

Rationale: To support the asset in meeting the facility's life expectancy.

5.2.2.2 Outside battery limits

1. Grass or other ground cover plants shall not be used as erosion protection in flare areas, major pipe tracks, pump pits, or other specified areas susceptible to fire hazards.

Rationale: Due to the fire risk during seasons when vegetation dies or go dormant.

2. Refer to (5.2.2.1) for the requirements for concrete panels.
3. Drainage points or weep holes as specified in (5.2.2.1, Item 6) shall be provided for flexible erosion protection systems.
4. Concrete linings shall be installed with a reinforcement mesh of 7 mm (1/4 in) steel wire spaced at 200 mm x 200 mm (8 in x 8 in).

Rationale: The reinforcement provides additional stability.

5.2.3 Off-plot areas

1. For areas not covered by (5.2.2), grass or ground cover plants may be used.
2. If grass is specified, the use of multiple slow growing varieties should be provided.

Rationale: Slow growing grass will require less maintenance.

5.3 MATERIALS

1. Erosion protection materials shall be consistent with ASTM Volume 04.03, or equivalent.
2. Refer to DEP 34.19.20.31-Gen. for the requirements of reinforced concrete structures.

5.4 CONSTRUCTION

1. Slopes shall be compacted and trimmed before erosion protection layers are installed to prevent local failure.

6. FENCING AND GATES

6.1 GENERAL

This section covers only the minimum requirements for permanent fences and gates. It does not cover special security measures that might be required such as cameras, security guards, temporary fencing and other specialty fencing systems.

1. The following applicable standards shall be used to supplement the requirements of (6.2) to (6.7):
 - a. ASTM A392;
 - b. ASTM A491;
 - c. ASTM D1557;
 - d. ASTM F552;
 - e. ASTM F567;
 - f. ASTM F626;
 - g. ASTM F668;
 - h. ASTM F1043;
 - i. BS 1722.

6.2 SECURITY RISK ASSESSMENT

1. For green field developments and extensions of existing facilities , fences and gates shall be erected within the site limits in compliance with the requirements provided by the security risk assessment.

The security risk assessment classifies the site according to the risk categories listed in Table 7.

Rationale: The security risk assessment identifies the category of risk the site is exposed to in accordance with those described in Table 7.

Table 7 Risk categories

Risk	Description
Insignificant Risk	There is a minimum level of crime and a need for basic personal and company security precautions. Generally, an effective public security infrastructure is in place.
Low Risk	There are frequent incidents of petty crime, limited possibility of activism or terrorism, and generally good quality public security infrastructure. Basic personal and company security are required.
Medium Risk	There are frequent incidents of crime, violence perpetrated by criminals, terrorists or guerrillas, and the possibility of small-scale internal unrest. Generally limited effectiveness of the public security infrastructure. Attention has to be paid to good personal and company security precautions.
High Risk	There is frequent petty, serious, and organized crime. A serious terrorist or guerrilla problem, with the possibility of medium-scale internal unrest with the threat of internal armed conflict. Generally poor quality public security infrastructure. Stringent personal and company security precautions are required.
Critical Risk	Security authorities no longer have control of law and order. Personnel and the company have to maintain their own highest possible levels of protective measures.

6.3 SELECTION OF FENCING

The security risk in the area and the suitability of locally available materials will govern the appropriate selection of fencing for a particular site.

1. The fencing system may consist of the following components:
 - a. all posts for line, corner and end have a buried concrete footing and inclined top;
 - b. bracing;
 - c. chain link fence fabric;
 - d. barbed wire;
 - e. tension wire;
 - f. gates.

6.4 FUNCTIONAL REQUIREMENTS

6.4.1 Fences

1. A system of fences and gates shall be erected close to and entirely within the site limits.

2. The fencing and gate system shall be designed to prevent unauthorised entry into the site.
The fencing system might consist of a single or double fence as required by the site security assessment.
3. The fencing system shall be a minimum of 2.4 m (7 ft - 10 in) in height.
The need for weld mesh, expanded metal or metal palisade fencing material will be specified in the requisition.
4. The top of the external boundary fence shall have a 45 degrees outward inclined section of 0.7 m (2.33 ft) length with anti-climb topping to 2.9 m (9.5 ft) overall.
5. The top of the outward inclination shall be within the property limits.
6. Where natural elevation, gradient, or man-made structures compromise the height of a 2.9 m (9.5 ft) perimeter or critical area fence, additional fencing material shall be added to ensure an effective overall height of 2.9 m (9.5 ft) is maintained.
7. Anti-climb topping of the fence system shall be one of the following:
 - a. 3–6 strand barbed wire top-guard constructed of 9 gauge or heavier wire angled out and up at a 45 degree angle;
 - b. concertina wire;
 - c. razor-ribbon.
8. The fence system shall facilitate discovery by plant security of any trespassing.
9. A clear line of sight shall be established along the length of the fence with hills and gullies along the fence line being levelled off as far as reasonably practicable.
10. Fences shall have a minimum number of bends.
11. The fence system shall not inhibit the natural dispersion of gas leakages on site.
Rationale: The fence cannot impede natural air movements that the site depends on to disperse any accidental release or provide ventilation.
12. At locations where fencing cannot be anchored to the base, gaps between the bottom of the fence and grade shall not exceed 5 cm (2 in).
Burial includes direct burial of the mesh in earth or embedding or anchoring the mesh to an embedded concrete kerb.
13. Storm drains, culverts, pipelines, and utility tunnels, in excess of 620 cm² (96 in²) that pass through or under the perimeter fence, shall be fitted with additional security screening or bars to prevent intrusion.
14. The arrangement and design of the egress points in the fencing and gate system shall accommodate all of the following:
 - a. the population and security logistics during each shift change;
 - b. all visitor types and designations;
 - c. all emergency responses;
 - d. all deliveries;
 - e. all shipments from the site;
 - f. all site operations and maintenance requirements.

6.4.2 Gates

1. Gates, if specified, shall be an integral part of the fence.
2. Gates, if specified, shall be of a similar design with respect to height and strength.

Rationale: Gates need to meet the operational, administrative and security requirements. In some cases, this can include the addition of anti-vehicular devices like hydraulic bollards or ramps integral with the gate or gate automation that will involve other disciplines.

3. Gates shall be of the swinging type.

6.5 DESIGN REQUIREMENTS

6.5.1 General

1. Prior to the erection of fences, site preparation shall fulfil the following requirements:
 - a. embankments do not extend beyond the site boundary;
 - b. embankments are stable and have a slope of maximum 1 vertical to 1.5 horizontal;
 - c. erosion protection applied as specified in (5);
 - d. embankment fill material placed at a minimum dry density of 85 % modified AASHTO (ASTM D1557 or AASHTO T180);
 - e. embankments have a minimum shoulder of 0.5 m (1.67 ft) between the fence foundation and the outside embankment slope to permit inspection;
 - f. embankment slopes resulting from site preparation works allow a clear view from the patrol road or maintenance track towards the area outside the fence.

6.5.2 Fence design

1. The fence shall be designed to withstand any load that may be exposed to within its design life (e.g. gale force wind loads, thermal movement from extreme climatic conditions).
2. The extent of support at the base of the fence shall be based on the local environmental and climatic conditions as well as the requirements necessary to address the risks identified by the assessment in (6.2) and other considerations (e.g. national requirements).
3. Ground beams and anchors should be used to prevent the bottom of the fence from lifting up.
4. The concrete footing of a fence post shall extend at least 0.6 m (2 ft) into the ground.
5. The concrete footing of a fence post shall have at least the same weight as the post with a minimum of 25 kg (55 lb).
6. Metal parts, except the chain link fence fabric, shall not be in contact with the soil.

6.5.3 Drainage crossings

1. A removable bar screen barrier shall be installed at crossings of the fence with water courses or other drainage channels for security reasons to prevent unauthorized entry.

6.6 MATERIALS

Galvanized steel, aluminium and concrete are acceptable material for fences. PVC-coated chain link fence fabric and tension wire may be used.

1. Fence materials shall be compatible with the climatic conditions, soil conditions and plant atmosphere.
2. The material shall be selected to mitigate against contact corrosion when assembled.

6.7 TIMING OF CONSTRUCTION

1. The construction of permanent fences and gates may precede all other construction activities.
2. If permanent fences and gates do not precede all other construction activities, a temporary fencing system shall be placed.

7. REFERENCES

In this DEP, reference is made to the following publications:

Unless specifically designated by date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.

SHELL STANDARDS

DEP 00.00.05.80-Gen.	DEP feedback form
DEP 00.00.06.06-Gen.	Standard drawings index
DEP 32.37.20.10-Gen.	Instrument signal lines
DEP 33.64.10.10-Gen.	Electrical engineering design
DEP 33.64.10.33-Gen.	Electromagnetic compatibility (EMC)
DEP 34.11.00.11-Gen.	Site preparation and earthworks including tank foundations and tank farms
DEP 34.11.00.12-Gen.	Geotechnical and foundation engineering - Onshore
DEP 34.14.20.31-Gen.	Drainage systems and primary treatment facilities
DEP 34.19.20.31-Gen.	Reinforced concrete structures
DEP 80.00.10.11-Gen.	Layout of onshore facilities
DEP 80.47.10.31-Gen.	Fixed, portable and automotive active fire protection systems and equipment for onshore facilities

STANDARD DRAWINGS

S 19.001	Electrical and instrument cable trenches in concrete paved areas.
S 19.002	Cable routing in unpaved, brick-paved or tiled areas and crossing roads.

AMERICAN STANDARDS

UFC 3-320-06a	Concrete Floor Slabs on Grade Subject to Heavy Loads Issued by the Department of Defense: USA
	Port and Industrial Pavement Design with Concrete Pavers Issued by the Interlocking Concrete Pavement Institute (ICPI)
AASHTO T180	Standard method of test for moisture-density relations of soils using a 4.54-kg (10-lb) rammer and a 457-mm (18-in) drop
AASHTO GDPS-4	Guide for design of pavement structures <i>Issued by: American Association of State Highway and Transportation Officials</i>
ACI 316R	Recommendations for construction of concrete pavements and concrete bases
ACI 325.9R	Guide for construction of concrete pavements and concrete bases <i>Issued by: American Concrete Institute</i>
ACI 350	Code Requirements for Environmental Engineering Concrete Structures and Commentary

ASTM A392	Standard specification for zinc-coated steel chain-link fence fabric
ASTM A491	Standard specification for aluminium-coated steel chain-link fence fabric
ASTM D1557	Standard test methods for laboratory compaction characteristics of soil using modified effort (56,000 ft-lbf/ft ³ (2,700 kn-m/m ³))
ASTM F552	Standard terminology relating to chain link fencing
ASTM F567	Standard practice for installation of chain-link fence
ASTM F626	Standard specification for fence fittings
ASTM F668	Standard specification for Polyvinyl Chloride (PVC), Polyolefin and other polymer-coated steel chain link fence fabric
ASTM F1043	Standard Specification for Strength and Protective Coatings on Steel Industrial Fence Framework
ASTM D1883	Standard test method for California Bearing Ratio (CBR) of laboratory-compacted soils.
ASTM D1195	Standard test method for repetitive static plate load tests of soils and flexible pavement components, for use in evaluation and design of airports and highway pavements.
ASTM D2844	Standard test method for resistance R-value and expansion pressure of compacted soils.
ASTM 04.03	Road and paving materials; vehicle-pavement systems
PCA IS059P	Joint Design for Concrete Highway and Street Pavements <i>Issued by: Portland Cement Association (PCA)</i>

BRITISH STANDARDS

BS 1722	Fences
	Design Manual for Roads and Bridges of the Highway Agency (UK)

EUROPEAN STANDARDS

EN 1338:2003	Concrete paving blocks - Requirements and test methods
EN 1992-3	Eurocode 2 - Design of concrete structures - Part 3: Liquid retaining and containment structures

OTHER REFERENCES

Port and Industrial Pavement Design with Concrete Pavers, 2nd edition, by J. Knapton
Issued by: Interlocking Concrete Pavement Institute